

The Restoration of Elk (*Cervus elaphus*) in Ontario, Canada: 1998–2005

Rick Rosatte,^{1,2} Joe Hamr,³ Jim Young,⁴ Ivan Filion,³ and Howard Smith⁵

Abstract

In 1997, a plan to restore Elk (*Cervus elaphus*) to Ontario was approved by the provincial government. The objective of the Ontario elk restoration program, a multipartnered collaboration, was to restore a species that had been extirpated from the province during the 1800s. During 1998–2001, 460 elk were acquired from Elk Island National Park, Alberta, for release in four areas of Ontario. As greater than 90% of the elk were radio collared, monitoring provided detailed information on the dynamics of the four populations. Comprehensive research projects using graduate students were implemented to determine the environmental impact of releasing elk in Ontario. Those studies are in progress or have been completed and include the effect of wolf predation on restored elk, white-tailed deer and elk resource overlap, the development of genetic profiles for elk, and solutions for elk/human conflicts. Mortality of the released elk averaged 41% (190/460) during 1998–2004 with annual mortality generally

declining over time in each release area. The primary causes of elk mortality included wolf predation (25% of mortalities), illegal shooting (13%), stress-related emaciation (13%) (partially due to the stress of relocation), bacterial infections (7%), and collisions with vehicles (6%). Productivity has been high in one of the release areas with 24–65% of the cows being observed with calves during late winter surveys. However, productivity has been low in two of the northern release areas due to a variety of factors including wolf predation. In some areas, dispersion of elk appeared to be related to the length of time animals were kept in pens prior to release. The precalving population estimate for Ontario in March 2004 was 375–440 elk. A comprehensive program review was conducted in 2003/2004 that included recommendations relating to the future management of elk in Ontario.

Key words: *Cervus elaphus*, elk, Ontario, restoration.

Introduction

History of Elk Restoration in North America

Elk (*Cervus elaphus*), also known as wapiti, were historically distributed across much of the United States and parts of Canada (Murie 1951; Bryant & Maser 1982; O’Gara & Dundas 2002; Peek 2003). Although accurate methods of determining elk population size were not available at the time, Seton (1927) estimated that there were about 10,000,000 elk in North America prior to the immigration of Europeans (Bryant & Maser 1982). By the early 1900s, elk populations across North America were severely depressed (estimated at about 100,000) due to unregulated meat hunting, severe winters in some areas, and eradication due to competition with livestock and crop damage. In simple terms, elk were exterminated because of competition for settled land (Bryant & Maser

1982). In fact, two subspecies, Meriam (*C. elaphus merriami*) and Eastern (*C. elaphus canadensis*) of elk were extinct, one, Tule (*C. elaphus nannodes*), had been eradicated from most of its range, and three, Rocky Mountain (*C. elaphus nelsoni*), Roosevelt (*C. elaphus roosevelti*), and Manitoban (*C. elaphus manitobensis*), were near extinction by the early twentieth century (Bryant & Maser 1982). Fortunately, National Parks were created in the United States and Canada during this period to protect elk and other natural resources, and by the 1970s, there were an estimated 500,000 elk in North America (Bryant & Maser 1982).

Numerous reintroductions or supplementations were initiated across North America in the early part of the twentieth century in an attempt to rebuild elk populations that were decimated during the 1800s (Bryant & Maser 1982; O’Gara & Dundas 2002). Some of those efforts were successful, whereas others failed for a variety of reasons (Severinghaus & Darrow 1976; Bryant & Maser 1982). The majority of elk for those restoration attempts were acquired from Yellowstone National Park and Jackson Hole, Wyoming. Recent (1995–2001) reintroduction attempts have occurred in Kentucky, Ontario, North Carolina, Tennessee, and Wisconsin (O’Gara & Dundas 2002; Rosatte et al. 2002; Larkin et al. 2003, 2004). Due in part to restoration, elk now occupy more land across North America than at any time since the early twentieth century with an

¹ Ontario Ministry of Natural Resources, Wildlife Research and Development Section, Trent University, Science Complex, PO Box 4840, Peterborough, Ontario, Canada K9J 8N8.

² Address correspondence to R. Rosatte, email rick.rosatte@mnr.gov.on.ca

³ Northern Environmental Heritage Institute, Cambrian College, 1600 Barrydowne Road, Sudbury, Ontario, Canada P3A 3V8.

⁴ 24 Karen Drive, Lindsay, Ontario, Canada K9V 5V5.

⁵ Ontario Ministry of Natural Resources, Wildlife Section, 300 Water Street, Peterborough, Ontario, Canada K9J 8M5.

estimated population of 782,000 to 1 million, depending on the reference (Peek 1999, 2003; O'Gara & Dundas 2002).

History of Elk in Ontario

Once native to Ontario, elk were extirpated by the late 1800s as a result of several factors including unregulated hunting, market hunting, and the eradication of elk on lands targeted for human settlement and agriculture. There have been several attempts to restore elk to the province, including one in the early 1930s where 24 Rocky Mountain elk from Alberta were released in Ontario. Additional translocations during the 1930s brought the total number of elk released in the province to about 200. By the mid-1940s, the Ontario elk population was estimated at about 300. Unfortunately, most of these animals and their offspring were subsequently killed by provincial officials due to unfounded concerns that the elk were infecting cattle with the giant liver fluke (*Fascioloides magna*). It is estimated that about 1,000 elk were killed during the late 1940s and early 1950s. However, two small herds of elk somehow managed to survive in the Burwash/French River area of Central Ontario and have been protected since 1980. In 1996, provincial elk numbers were estimated at about 50–60 animals. During the mid to late 1990s, a plan was drafted to supplement that population and restore elk to other areas of Ontario.

Ontario's Elk Restoration Plan

The plan for the restoration of elk in Ontario (Bellhouse & Broadfoot 1998) identified six broad geographic areas in Ontario that contained suitable elk habitat (Fig. 1) and recommended that up to 200 animals be introduced to each release area to ensure the establishment of viable

self-sustaining populations. Ontario's elk restoration initiative involved many partner groups, volunteers, and sponsors including Cambrian College, Elk Island National Park (EINP) (Parks Canada), French River Resorts Association, Ontario Federation of Anglers and Hunters, Ontario Fur Managers Federation, the Rocky Mountain Elk Foundation, Safari Club International (Ontario Chapter), Sault College, Trent University, Laurentian University, Lakehead University, and the Ontario Ministry of Natural Resources (OMNR). Local Implementation Committees (LICs) to support fund raising and monitoring have been established in the four areas where elk have been introduced.

Methods

Source of Elk for Ontario's Elk Restoration Effort

EINP near Edmonton, Alberta, was the source of elk for Ontario's current (1998–2004) restoration effort. EINP operates one of the most rigorous disease management programs in North America in cooperation with the Canadian Food Inspection Agency (CFIA). The park's boundary is completely fenced preventing the animals from dispersing naturally and this, along with a lack of predators, results in a surplus of elk that can be removed annually. Individual elk were assessed, sexed and aged, treated for parasites and diseases, ear tagged, and fitted with radio collars using the process outlined in Rosatte et al. (2002).

Transportation, Animal Holding, Release, and Monitoring of Elk

Elk were shipped to Ontario in commercial transport trailers, Rocky Mountain Elk Foundation Canada (RMEFC) gooseneck trailers, or commercial stock trailers and were provided hay and water during all shipments. Shipment time from EINP to Sudbury (except during 1998) was 40–46 hours, to Blind River 48 hours, to Kenora 24 hours, and to Bancroft 48–58 hours, with weather and road conditions determining the exact timing.

Upon arrival in Ontario, the elk were placed in holding pens to recuperate from the trip and to habituate to their new surroundings. The length of the holding period varied from an unplanned immediate release (hard release) to 4–10 days (semisoft release) to 6–16 weeks (soft release). Once released, elk were monitored using radiotelemetry. Attempts were made to submit elk mortalities to the Canadian Cooperative Wildlife Health Centre, Guelph, Ontario, for postmortem to determine the cause of death. To date (2006), elk have been released within four of the six areas identified in the plan to restore elk in Ontario: (1) Nipissing/French River (NFR); (2) Haliburton Highlands (later changed to Bancroft/North Hastings [BNH]); (3) Lake of the Woods (LOW); and (4) Lake Huron North Shore (LHNS) (Fig. 1). Further shipments have not occurred due to the threat of chronic wasting disease (CWD).



Figure 1. Potential elk release areas in Ontario.

Elk Research Programs

In 1999, the Elk Restoration Network together with Ontario's elk technical experts assessed Ontario's research needs and priorities and designed a research program with the following objectives:

- To determine the environmental and interspecific impact of restoring elk to Ontario.
- To initiate research projects that will assist in managing elk herds in Ontario to attain maximum growth potential and survival.
- To initiate a series of research projects that will provide data for the development of a management plan for elk in Ontario.

The most feasible approach to implement research programs was by employing graduate students. Since 2000, a total of 11 graduate research programs focusing on the restored elk have been initiated at several universities in Ontario including Laurentian, Trent, Lakehead, and Guelph. Investigations included the population dynamics of introduced herds, the potential for competition between elk and white-tailed deer, elk habitat signatures, long-term population censusing methods, development of a post-release elk dispersion model, the impact of predation on released elk, genetic profiling, and nuisance behavior mitigation (Fig. 2). In late 2003, a 5-year review of Ontario's elk restoration and research program was completed. This document identified future challenges and opportunities.

Results

Numbers, Sex, and Age Classes of Animals Shipped

Between January 1998 and February 2001, 460 elk were transported from Alberta to Ontario. The sex and age



Figure 2. R. Rosatte and J. O'Donnell replacing the radio collar on an immobilized cow elk near Bancroft, Ontario. (Photo by Ryan Cavanaugh.)

distribution for these elk is shown in Table 1. An approximate ratio of 2.5 cows to 1 bull was requested to maximize productivity and herd growth. However, this was not always possible due to the sex and age distribution of elk that were captured at EINP during any given year.

Mortality

Total accumulated mortality among the 460 elk that were transported to Ontario was 41% (190/460) between 1998 and 2004 (Table 2). However, 17 of those mortalities occurred in the holding pens prior to release. Most of these mortalities were thought to be stress related (i.e., pneumonia, injury related infections, etc.). Wolf predation was high for the first two (1998/1999) shipments to the NFR elk release area (ERA) (Table 2). Fourteen of 69 animals released in the NFR area in mid-January, 1999, were lost by the end of April as a consequence of post-release stress. Mortality due to meningeal worm infection may become a factor in the future in the BNH area as several elk (59% of a sample) have been suspect for *Parelaphostrongylus tenuis*. About 82% of white-tailed deer fecal samples were positive for *P. tenuis* larvae (McIntosh et al. 2006) (Fig. 3).

Cumulative calf mortality averaged 38% (50/130) for all years and all areas (Table 3). Mortality for bulls in all areas averaged 33% (28/84) and cow mortality was 46% with yearly as well as area-specific differences (Table 3). The 28% total mortality figure for LOW during 2000 (Table 2) is somewhat biased as four of the mortalities were noncollared animals and so may not reflect the true degree of mortality. Once elk had been on the landscape for a few years in Ontario, annual mortality tended to decrease (Fig. 4).

Productivity

The percentage of cows with calves during late winter surveys has varied greatly since the inception of the elk restoration program with a high of 65% in the BNH area during 2004 to a low of 0% in the NFR area in the winter of 2001.

NFR Area. In the winters (February–March) of 1999 and 2000, only 10–20% of recently introduced adult cows in the NFR area were accompanied by calves. That included sightings of 6 calves during 1999 and 12 during 2000. A December 2000/January 2001 calf survey showed that over 30% of the adult cows in the NFR elk population (both resident and restored elk) had calves (22 in total) (Table 4). Wolf-killed elk calves, born in NFR, were observed in the winters of 1999, 2000, and 2001. Elk remains were found in the stomachs of four out of six wolves snared by trappers in the core of the NFR elk range during December 2000. During an April 2002 survey, only 1 calf was observed per 35 cows. Two calves were found dead on a railway line and wolves killed at

Table 1. Number of elk shipped to Ontario from EINP, Alberta during 1998–2001.

<i>Date Shipped</i>	<i>Release Area</i>	<i>Radio-Collared Elk</i>	<i>Elk Shipped</i>	<i>Elk Released</i>	<i>Adult/Yearling Bulls</i>	<i>Adult/Yearling Cows</i>	<i>Male Calves</i>	<i>Female Calves</i>
March 1998	NFR	47	47	40	14	33	0	0
January 1999	NFR	66	69	69	7	42	6	13
March 2000	NFR	35	40	40	0	20	7	13
February 2001	NFR	25	26	23	1	10	7	8
January 2000	BNH	70	70	70	14	36	10	10
January 2001	BNH	50	50	50	13	20	6	11
January 2000	LOW	30	60	60	13	33	8	6
February 2001	LOW	43	48	44	13	21	9	5
December 2000	LHNS	50	50	47	9	30	8	3
Total elk 1998–2001	ALL AREAS	416	460	443	84	245	61	69*

*Sex was unknown for one calf.

least one during the late winter of 2002. This combined with the 2002 survey data indicates a high mortality rate for calves in the NFR ERA. During the winter of 2003, there was little accumulated snowfall in the NFR area with strong ice conditions and no snow crust. The late winter survey showed 30% of the observed cows were accompanied by calves (Table 4). During the winter of 2004, there was once again little accumulated snowfall in the NFR area, and the early winter survey showed a promising 37% of observed cows with calves. However, this dropped to 23% in the late winter survey. There was some level of predation during the winter of 2004 with three elk calves confirmed as wolf kills.

BNH Area. About 14 calves were confirmed in the BNH ERA during the summer/fall of 2000. A total of 25 cow elk were observed during a helicopter survey on 20–23 February 2001 in the Bancroft ERA. Ten of those cows (40%) had calves (Table 4). During an aerial survey in the Bancroft ERA on 6–9 March 2002, 14 calves were

observed as well as 37 mature cow elk. The winter of 2002/2003 was severe with high snow depths—although there had been many public reports of calves during the previous summer and fall, the aerial survey in March 2003 detected 24% of the observed cows with calves (9/37) (Table 4). The 2003/2004 winter survey resulted in record numbers of calves sighted in the BNH area. A total of 29 calves were observed in the company of 58 cows (50%) during the January survey; the March survey showed even better results as 31 (65%) of the 48 observed cows had calves (Table 4).

LOW Area. Three calves with 10 cows (30%) were observed in the LOW area during 2001. Only 4 calves and 23 cows (17%) were observed during an aerial survey in March 2002 and 4 calves with 22 cows (18%) during the March 2003 survey. This trend of low calf numbers continued during 2004 when surveys were conducted in January and February, and only three calves were observed in the company of 20 cows (15%) (Table 4).

Table 2. Causes of mortality of elk released in four areas of Ontario from 1998 to 31 March 2004.

<i>Area (ERA)</i>	<i>Year Elk Collared</i>	<i>Total^a Mortality % (n)</i>	<i>Emaciation % (n)</i>	<i>Wolf Predation % (n)</i>	<i>Injury % (n)</i>	<i>Shot^b % (n)</i>	<i>Drowning % (n)</i>	<i>Road Kill % (n)</i>	<i>Bacteria Infection % (n)</i>	<i>Other^c % (n)</i>
NFR	1998	57 (27/47)	0	19 (9)	11 (5)	2 (1)	4 (2)	0	13 (6)	9 (4)
NFR	1999	67 (46/69)	23 (15)	29 (19)	2 (1)	3 (2)	4 (3)	0	2 (1)	8 (5)
NFR	2000	38 (15/40)	3 (1)	20 (8)	3 (1)	0	8 (3)	5 (2)	0	0
NFR	2001	50 (13/26)	0	19 (5)	0	0	0	8 (2)	0	23 (6)
BNH	2000	37 (26/70)	10 (7)	0	4 (3)	8 (4)	0	6 (4)	0	11 (8)
BNH	2001	34 (17/50)	0	2 (1)	0	10 (5)	0	4 (2)	0	18 (9)
LOW ^d	2000	30 (18/60)	0	5 (3)	0	13 (8)	2 (1)	0	0	10 (6)
LOW	2001	33 (16/48)	0	6 (3)	0	4 (2)	4 (2)	2 (1)	8 (4)	8 (4)
LHNS	2001	24 (12/50)	2 (1)	0	4 (2)	6 (3)	0	2 (1)	4 (2)	6 (3)
All	98/01	41 (190/460)	5 (24)	10 (48)	3 (12)	5 (25)	2 (11)	3 (12)	3 (13)	10 (45)

^aTotal mortality is accumulated mortality from release year to 31 March 2004. Seventeen of the mortalities (10 in NFR, 4 in LOW and 3 in LHNS) occurred in the holding pens prior to release.

^bShot refers to elk that were intentionally poached, accidentally shot, or shot in the United States (via permit).

^cOther sources of mortality include bear predation (2), accident (fell over cliff) (1), unknown (20), train (4), tumor (1), hypothermia (1), capture myopathy (1), euthanized (3).

^d2000 LOW mortality was based on five radio-collared mortalities and four uncollared mortalities (30 of 60 elk had radio collars).



Figure 3. Restored elk and white-tailed deer now share range in Ontario with risk of transmission of meningeal worm from deer to elk. (Photo R. Rosatte.)

LHNS Area. In the LHNS area, during the late winter calf surveys, eight calves were observed during 2002 and six in 2003. The 2003 survey was carried out after the capture and transfer of a cow elk and three calves to the NFR area. Thus, calf survival has been consistent for the past two winters at 32 and 33% of cows with calves. The 2004 winter survey results are very consistent with 33% of cows accompanied by calves (7/21) (Table 4).

Population Estimates and Trends

Elk population estimates were determined by counting the number of marked (radio collared) and unmarked elk observed during aerial surveys and through input of that data into a population model (e.g. modified Petersen model). As of 31 March 2004, researchers estimated the provincial elk precalving population at between 375 and 440 animals: NFR, 110–130; BNH, 170–200; LOW, 35–45;

LHNS, 60–65. The NFR population showed decreases following the releases of elk between 1998 and 2001. However, during 2003/2004, the population has shown growth of about 10%. The BNH population has shown steady growth and there are now many thriving subherds located throughout the release area. The LOW population has continued to show slow decline. The high mortality due to poaching coupled with the low calf survival/recruitment are thought to be the principal factors. The LHNS population has shown steady growth since the initial release in 2001.

Elk Dispersion Following Release

Elk dispersion following release from the holding pens appeared to be related to the length of time they were held prior to release (Table 5). However, one cannot discount the effect of factors such as habitat, predators, roads, and human disturbance on elk dispersion. Elk that were held for 6 weeks or longer, as during the 2000 NFR release, tended to remain in the vicinity of the holding pens. In fact, most of those elk are still (2005) within 5 km of the release site. Elk that experienced a hard release as in the 2000 BNH unintentional release tended to disperse over a much wider area. In fact, the BNH 2000 released elk spread over a 27,000 km² area. Only about 50% of the remaining 53 elk were within 20 km of the release site during 2001. Some animals (about 10) moved 100–140 km from the release site. Some elk (four) from the 2000 LOW release have ventured into Minnesota (about 70–90 km from the release site) and at least six elk dispersed to the Fort Frances/Rainy River area (about 40–60 km from the release site); however, most of the elk are ranging within 20 km of the release site. Elk movements (two) of up to 160 km (to Algoma Mills and to Parry Sound) have been noted for the 1998 (1.5 years after release) and 1999 NFR introduced animals (held for 26 and 4 days, respectively).

The longest movement of a 2000 NFR animal (held for 6 weeks) was 50 km. During 2002/2003, animal movements

Table 3. Cumulative mortality per sex and age class and release year of elk at the four elk restoration areas in Ontario during 1998 to 31 March 2004^a.

Area ^b (ERA)	Release ^c Year	Adult Males % (n)	Adult Females % (n)	Male Calves % (n)	Female ^d Calves % (n)
NFR	1998	29 (4/14)	70 (23/33)	NA	NA
NFR	1999	0 (0/7)	69 (29/42)	100 (6/6)	85 (11/13)
NFR	2000	NA	50 (10/20)	14 (1/7)	31 (4/13)
NFR	2001	100 (1/1)	50 (5/10)	28 (2/7)	63 (5/8)
BNH	2000	36 (5/14)	33 (12/36)	40 (4/10)	50 (5/10)
BNH	2001	38 (5/13)	35 (7/20)	50 (3/6)	18 (2/11)
LOW	2000	31 (6/13)	27 (9/33)	0 (0/8)	33 (2/6)
LOW	2001	15 (5/13)	38 (8/21)	33 (3/9)	20 (1/5)
LHNS	2001	22 (2/9)	30 (9/30)	0 (0/8)	33 (1/3)
All	1998–2001	33 (28/84)	46 (112/245)	31 (19/61)	45 (31/69)

NA = not applicable.

^aActual mortality may have been higher during the 1999 and 2000 NFR program as three of the 1999 calves and five of the 2000 cows did not have radio collars. Similarly, during the 2000 LOW program 18 cows, 4 calves, and 8 bulls did not have radio collars.

^bAny known mortalities among uncollared animals are included in this table.

^cMortality for each "release year" is total mortality accumulated from the year of release to 31 March 2004.

^dThe sex of one calf was not recorded.

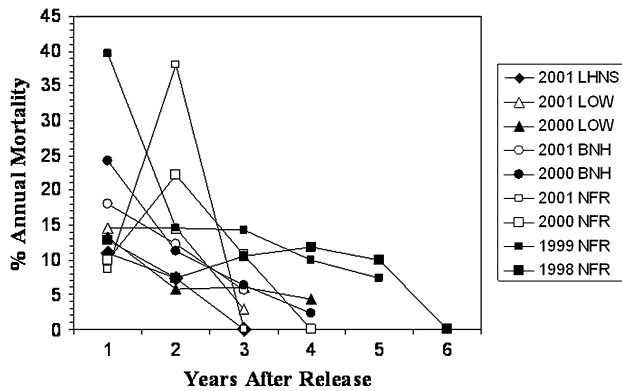


Figure 4. Postrelease annual mortality of elk reintroduced to four different areas of Ontario, Canada, during 1998–2001^a.

^aMortality is from the date of release until 31 December 2003. Total elk released in all areas is equal to 443, of which 416 were radio collared. Mortality includes only radio-collared elk. Total elk released in each area is as follows: LHNS, 47; LOW, 104; BNH, 120; NFR, 172. Only 50% (30/60) of the 2000 LOW elk were radio collared. Mortality increase in year 2 of 2000/2001 NFR elk release was due to wolf predation. Mortality does not include animals, which died in the release pen or that were euthanized due to injuries or bacterial infections prior to release.

continued to become more predictable and subherds became established in many areas. In the LOW area, there continued to be a scattering of individuals or small groups of two or three animals across the Rainy River and Fort Frances areas; the majority of animals continue to live within 10 km of the Cameron Lake release site. Nineteen animals were relocated from the Kynoch and Parkinson areas to a site north of Elliot Lake in the LHNS release area; subherds also exist in Parkinson and surrounding townships and at Neebish. Other individual or groups of two or three animals are scattered across the area with two cows having taken up residence in Michigan. There have been an increasing number of reports of sightings of individual and groups of elk in the Parry Sound area during the past 2 years. The relocation experience to date has been very successful with most animals staying near their relocation site and none reported to have caused any further problems.

Human/Elk Conflicts

In all of the release areas, elk have caused problems ranging from fence and crop damage to conflicting with red deer/elk farming operations. Solutions used to date range from trapping and relocating the elk to educating landowners on the benefits of elk. In 2003, nuisance elk guidelines were drafted to provide direction in dealing with nuisance situations.

Justification for Hunter Education

Since 2000, 25 of the 443 elk that were released in Ontario have been confirmed shot (as of 31 March 2004) (Table 2).

Table 4. Number of elk calves observed during surveys in four elk restoration areas of Ontario during 1998–2004.

Area (ERA)	Year	Number of Calves Observed	% (n) of Cows Observed with Calves
NFR	1997 (November)	4	40 (4/10) ^a
NFR	1998 (November)	4	36 (4/11) ^b
NFR	1999 (January)	5	55 (5/9)
NFR	2000 (February)	4	21 (4/15)
NFR	2000 (March)	6	26 (6/23)
NFR	2000 (December)	15	33 (15/46)
NFR	2001 (January)	6	38 (6/16)
NFR	2001 (April)	0	0 (0/14)
NFR	2002 (January)	5	21 (5/24)
NFR	2002 (April)	1	3 (1/33)
NFR	2002 (December)	10	24 (10/41)
NFR	2003 (March)	10	29 (10/34)
NFR	2004 (January)	19	37 (19/52)
NFR	2004 (March)	5	23 (5/22)
BNH	2001 (February)	10	40 (10/25)
BNH	2002 (March)	14	38 (14/37) ^c
BNH	2003 (March)	9	24 (9/37)
BNH	2004 (January)	29	50 (29/58)
BNH	2004 (March)	31	65 (31/48)
LOW	2001 (February)	3	30 (3/10)
LOW	2002 (March)	4	17 (4/23)
LOW	2003 (March)	4	18 (4/22)
LOW	2004 (Jan/Feb)	3	15 (3/20)
LHNS	2002 (March)	8	32 (8/25)
LHNS	2003 (March)	4	22 (4/18) ^d
LHNS	2004 (March)	7	33 (7/21)

^aNative animals, prerestoration/includes yearlings and adults.

^bSurvey included both resident and restored elk.

^cFive yearling cows were not included in the calculation.

^dThree calves removed from LHNS area to NFR in 2003. Assuming that at least two of these calves belonged to the observed cows, the percentage of cows that had surviving calves would be 33%.

A number of these animals were pregnant or had calves that may not have survived alone making the probable damage greater than the numbers indicate. There have also been at least six uncollared animals confirmed shot, four of these in the LOW area in 2003. There have also been several other unconfirmed reports of elk shootings, including nine in the LOW area. Data detailing the incidents were available for 22 of the elk that were shot. Of those, 13 (59%) were mistakenly identified by hunters as deer or moose. Seven (32%) elk were poached and two were shot in Minnesota as nuisance elk. In view of this, efforts have been made to increase hunter awareness of elk in Ontario, especially with respect to identification of elk.

Risk Assessment for CWD

Due to the threat of CWD, OMNR, in 2002, commissioned a risk assessment of continued elk reintroductions from the Canadian Cooperative Wildlife Health Center, Saskatoon. The report concluded that the risk of CWD being introduced to Ontario by elk from EIMP was low

Table 5. Length of holding period and dispersion of cow elk following release in four areas of Ontario during 1998–2001.

Elk Release Area	Release Date	Days Held in Pen	$\bar{X}$ (SD) Distance Moved (km)*	Shortest Distance Moved (km)	Greatest Distance Moved (km)	% (n) of Cow Elk Moving >50 km	% (n) of Cow Elk Moving <5 km	Total Cow Elk in Sample	Mortality from Date of Release to September of Release Year (%)
NFR	27 March 1998	27	16.7 (20.5)	1.0	84	4 (1/23)	44 (10/23)	23	8.6
NFR	15 January 1999	4	27.8 (24)	2.5	120	10 (3/31)	10 (3/31)	31	45.6
NFR	31 March 2000	43	5.1 (9.2)	0.5	50	(0/32)	91 (29/32)	32	11.1
NFR	14 April 2001	59	3.5 (2.8)	0.5	16.0	(0/23)	91 (21/23)	23	8.7
BNH	9 January 2000	0	31.7 (33.3)	4.0	140	16 (6/37)	5 (2/37)	37	20
BNH	14 April 2001	90	21.9 (19.1)	0.5	55	22 (4/18)	39 (7/18)	18	5.0
LOW	29 January 2000	10	37.7 (53.9)	1.0	200	36 (5/14)	36 (5/14)	14	7.1
LOW	9 March 2001	17	26.8 (28.3)	2.0	81	43 (6/14)	43 (6/14)	14	25
LHNS	8 April 2001	112	21.6 (27.5)	3.0	122	12 (3/26)	4 (1/26)	26	7.7

* Calculations include movements from the date of release to 30 September of the release year.

but recommended a conservative approach until more was known about the disease and the then recent cases in farmed cervids in Alberta. To ensure that Ontario elk are not infected with CWD, postmortems are done on all elk mortalities whenever feasible. Between 1998 and 2003, 73 heads from Ontario elk were tested for CWD under a pilot sampling program. Also, during the November 2002 deer hunt, OMNR sampled 151 hunter-killed deer from the Owen Sound area for testing at the Animal Health Lab. All tested samples were negative for CWD. In 2003, 454 white-tailed deer in the Lanark area also all tested negative. To date (January 2006), no cases of CWD in Ontario elk have been reported. As well, all elk mortalities at EINP receive a postmortem—again, no cases of CWD have been reported to date.

Discussion

The relocation of large mammals has been used globally as a technique to restore populations that have become threatened or endangered (Penzhorn 1971; Schmitt & Aho 1988; Griffith et al. 1993). In Ontario, Canada, elk were restored to four areas of the province during 1998–2005. However, elk populations appear to be increasing more dramatically in southern areas compared to the north. This is related to the degree of wolf predation, extent of illegal shooting of elk, as well as the impact of climatic conditions. EINP, Alberta (the source of the elk), biologists estimate that about 60–70% of the cow elk in the Park are with calves during each winter. This suggests that calf mortality in some areas of Ontario may be high due to a variety of mortality factors (i.e., wolf predation, high snow depths, poor ice conditions, etc.), although this may in part be due to the poor visibility of elk during aerial surveys. In the spring of 2003, researchers in the NFR area confirmed an incident involving bear predation on a newborn calf. Bears could thus be an important factor affecting elk recruitment and require further study. Wolf predation has continued to be an important factor in some winters in the NFR and also the LOW ERAs of Ontario. Poaching and illegal shooting have also been a concern especially in the LOW area. Poor calf recruitment in the LOW area may also have been due to a low bull:cow ratio. Relative to other restoration areas, there are few wolves in the BNH area, which may account for high calf survival compared to the LOW and NFR ERAs. However, it remains to be seen the degree of impact that meningeal worm infection will have on elk populations in southern Ontario (Samuel et al. 1992; Lankester 2001; Bellhouse & Rosatte 2005; McIntosh et al. 2006).

There are many things that must be considered to ensure that a restoration project is successful including the time of year for release, the stress of relocation, the duration the animals are held prior to release, and the potential for disease transmission (Collinson & Anderson 1984; Meltzer 1993; Woodford 1993; Short & Smith 1994). In this study, elk were held for considerable periods of time

prior to release in Ontario (i.e., soft release as opposed to a hard release). Those elk tended to remain closer to the release area and mortality has been lower than during years when elk were hard released or only held for a short period of time. However, it must be emphasized that these results are preliminary, and experiments on the length of holding time would need to be replicated before concrete conclusions can be made. Habitat type at the release site might also be a factor with favorable habitat enticing elk to remain in the vicinity of the release site. The time of year animals are released is also very important. For instance, a hard release in January, when there may be an abundance of snow and shortage of food may induce animals to move further and lose body reserves faster. In view of the results of this study, soft releases perhaps should be considered for the release of other large ungulate species in other jurisdictions (Hamr 2001; Bowyer et al. 2003).

In the past, illegal shooting has been a problem during large mammal restoration projects for species such as moose (Duvall & Schoonveld 1988). Similarly, in Ontario, illegal shooting of elk has been a problem in most of the ERAs. On several occasions, the infraction was due to misidentification of white-tailed deer for elk. Clearly, hunters in Ontario need to be educated to identify their target before shooting as there is no elk hunting season in the province. A number of initiatives have been undertaken to reduce this problem.

The issue of human/species interactions needs to be taken seriously with respect to elk management. Although the number of complaints received as a whole has been low, elk populations currently are well below the carrying capacity of the land. As elk populations flourish, especially in southern Ontario, human/elk interactions no doubt will escalate. Therefore, a well-designed nuisance elk management plan to deal with these interactions will be imperative.

A comprehensive disease management plan is a prerequisite for any mammalian restoration plan regardless of the species in question (Griffith et al. 1993; Meltzer 1993; Woodford 1993). All efforts must be made to prevent the spread of infectious diseases and parasites during species relocations (Nielsen & Brown 1988). In view of this, CWD has been reported in many U.S. states as well as in two provinces in Canada, including Alberta, the source for elk in this restoration program. As a result, further shipments of elk to Ontario will not occur until more is known about the probability of bringing CWD-infected elk from source herds in western North America. In addition, extensive sampling of deer and elk is occurring in Ontario to provide an early detection system for CWD and other diseases such as bovine tuberculosis.

Conclusions

The success of large mammal restoration programs has varied globally and has been dependent upon many factors (Collinson & Anderson 1984; Kojola et al. 1985; Borner 1988; Stuwe & Scribner 1989). The Ontario elk



Figure 5. Photo of restored elk herd near Bancroft, Ontario. (Photo by R. Rosatte.).

restoration program has met with great success primarily due to the collaborative, multipartnered approach to wildlife management. The first phase of the program involved acquisition and transport of elk to Ontario, followed by a comprehensive research program to determine the environmental impact of restoring elk to the province. A complete program review was undertaken in 2002 and 2003, and a number of recommendations were forwarded to the managing agency, the OMNR, for consideration. Phase two of the program, which will be the challenge for the future, will be to develop accurate methods to estimate elk density, so that elk can be managed in a sustainable manner for both consumptive and nonconsumptive uses, and to manage elk in such a way as to minimize conflicts between elk and humans (Fig. 5).

Implications for Practice

- Wild elk were restored to four areas of Ontario, Canada, proving that it is possible to restore extirpated populations of large ungulates. The methodologies used in this project could be applicable to other projects attempting to restore large ungulate populations.
- Attempts should be made to reduce stress during projects involving the movement of large ungulates such as elk. This should increase survival of the targeted species.
- Wolf predation and illegal shooting are limiting elk populations in two of the northern Ontario release areas. Only time will tell whether elk will flourish in Northern Ontario but winter severity most likely will be a contributing factor. Recent mild winters have contributed to the increased survival of elk in Ontario. The above factors should be considered when restoration of large ungulate populations are being considered in other areas.

- Meningeal worm infection may play some role in limiting elk populations where deer and elk ranges overlap, especially in southern Ontario. Attempts should be made to reduce the risk of transmitting this parasite to other susceptible large ungulate populations, e.g., the current range of meningeal worm is eastern North America and large ungulates such as moose (*Alces alces*) are susceptible. The risk of importation of CWD needs to be considered when restoring elk populations especially if the source herds are located in endemic areas. A risk assessment should be conducted prior to the initiation of any wildlife restoration project.
- A comprehensive disease management plan should be a requirement for elk restoration projects as well as any restoration project that involves the movement of large ungulates. This should minimize the impact of infectious diseases and parasites in the release area.

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